Computer Vision Project

Swarnim Kalra - RA2211026010205
Srijoni Chowdhury - RA2211026010210
Sayal Singh - RA2211026010218

CANNY EDGE DETECTION

Definition

- Canny Edge Detection is a multi-stage algorithm used to detect edges by finding areas of rapid intensity change in an image.

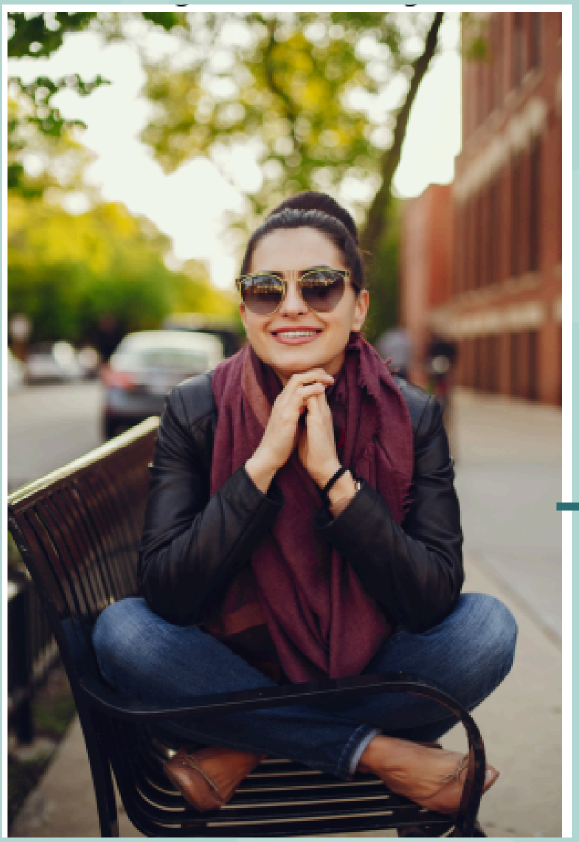
Purpose

- Accurate Edge Detection: Provides precise identification of true edges with minimal noise.
- Noise Reduction: Uses Gaussian filtering to smooth the image before detecting edges.
- Well-Connected Edges: Ensures edges are only one pixel wide and continuous.
- High Detection Rate: Detects both strong and weak edges effectively using dual thresholding.

Applications

- Object Detection
- Image Segmentation
- Medical Imaging

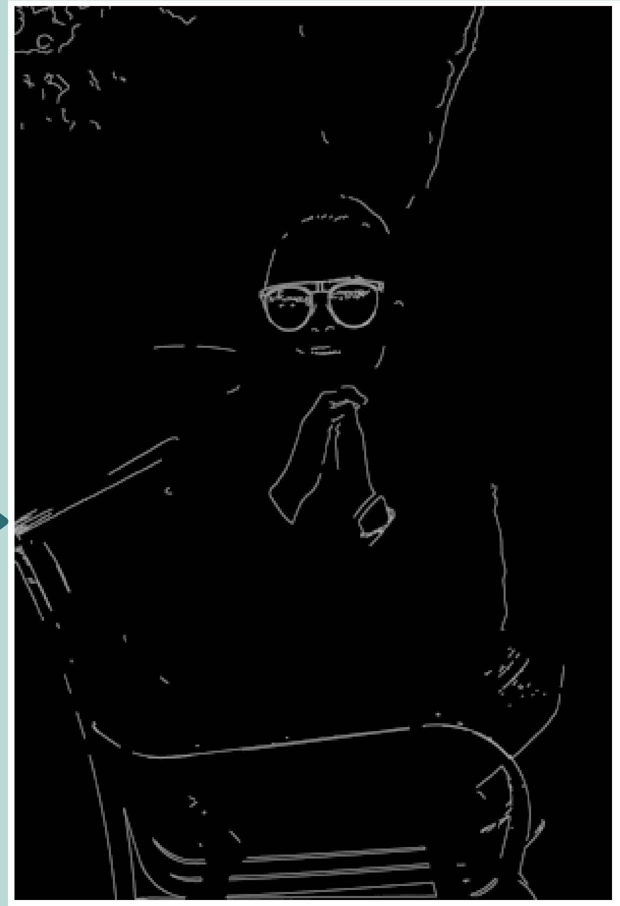
CANNY EDGE DETECTION



Original Image

Gaussian Blur Image

Gradient Calculation



Canny Output (Final Image)

CONTOUR DETECTION

Definition

- Contour detection identifies and traces the outline of objects in a binary image. Contours are curves joining all continuous points with the same intensity.

Purpose

- Shape Analysis: Understanding and analyzing the shapes and structures of objects.
- Object Recognition: Enables accurate identification of objects based on their outlines.
- Feature Extraction: Extracting geometric features like area, perimeter, and centroid.
- Efficient Image Segmentation: Simplifies complex images by outlining distinct object boundaries.

Applications

- Object Tracking
- Shape Matching
- Industrial Inspection

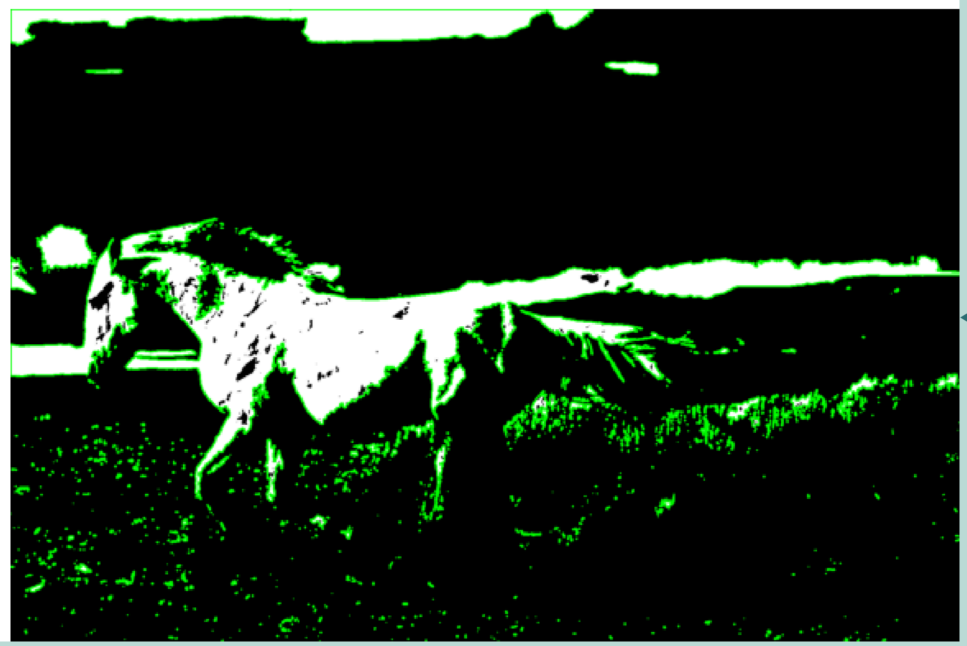
CONTOUR DETECTION

Original Image

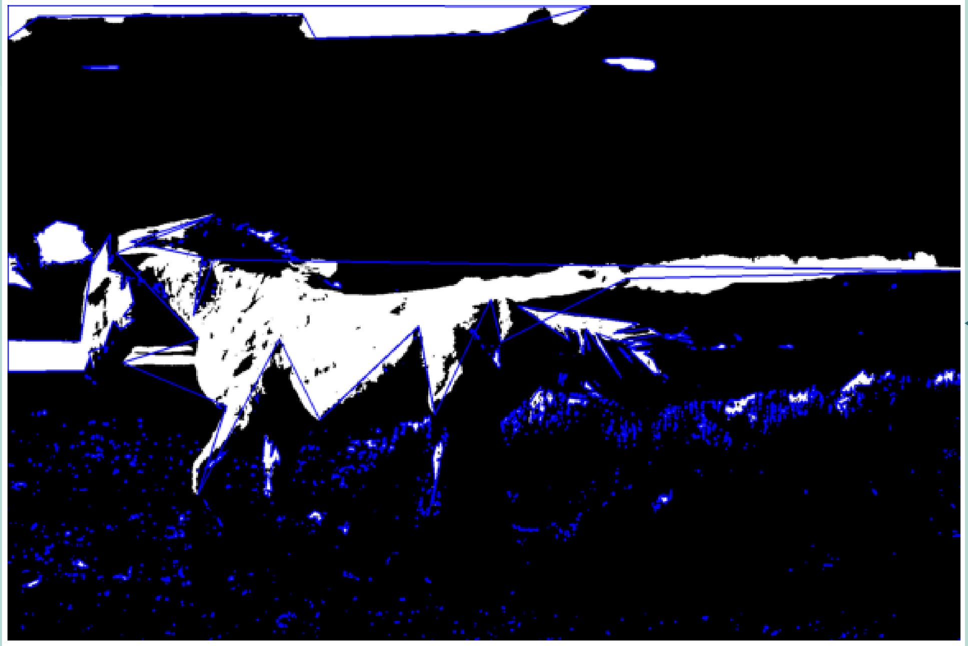
Gaussian Blurred Image

Binary Threshold Image

Detected Contours(1048)



Simplified Contours



HOUGH TRANSFORM

Definition

- The Hough Transform is a feature extraction technique used in image processing to detect geometric shapes like lines, circles, and ellipses.

Purpose

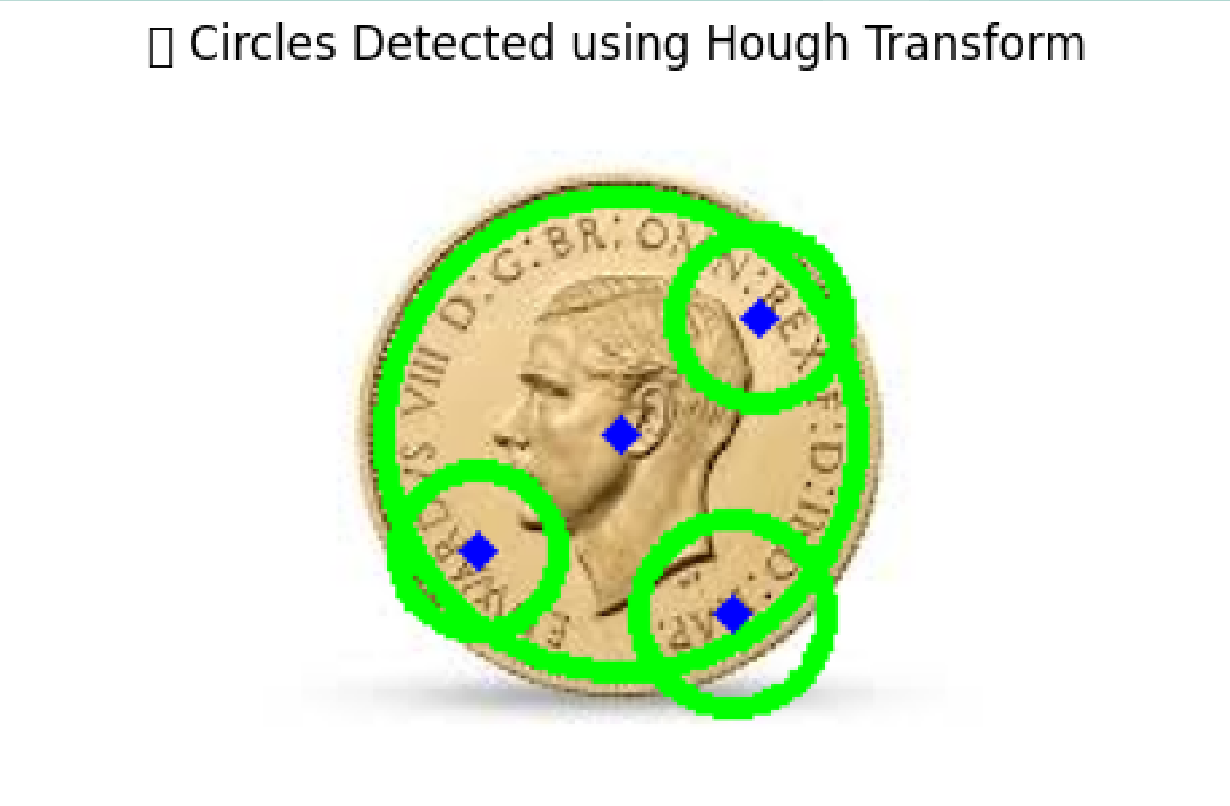
- To automatically detect circles in digital images, even if partially obscured or noisy.
- To provide robust shape recognition without relying on color or texture.
- To enable accurate geometric feature localization in computer vision tasks.

Applications

- Detecting coins, wheels, or circular signs in industrial vision systems.
- Iris or pupil detection.
- Medical imaging to identify circular structures like cells or tumors.

HOUGH TRANSFORM

Input Image



Circle detected using Hough Transform

METHODS FOR 3D VISION

Definition

- 3D Vision refers to techniques in computer vision that enable machines to perceive, analyze, and reconstruct the three-dimensional structure of a scene from one or more 2D images.

Purpose

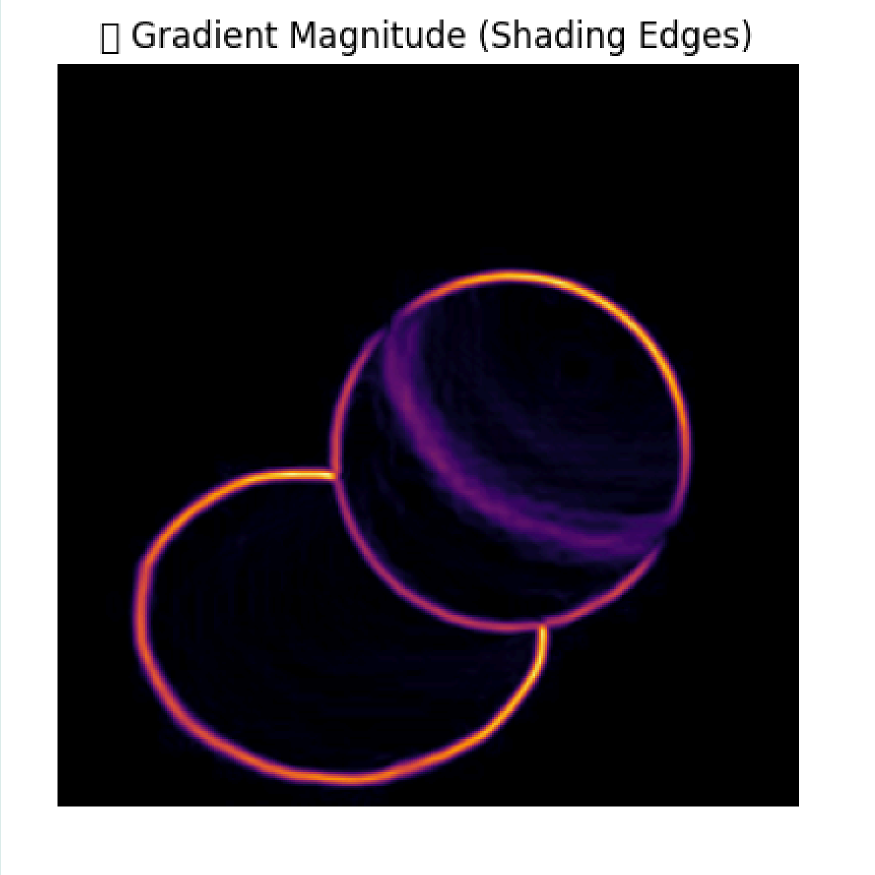
- To extract 3D shape and depth information from 2D images or video sequences.
- To reconstruct real-world scenes for visualization, measurement, or interaction.
- To enable accurate perception and tracking in robotics, AR/VR, and autonomous systems.

Applications

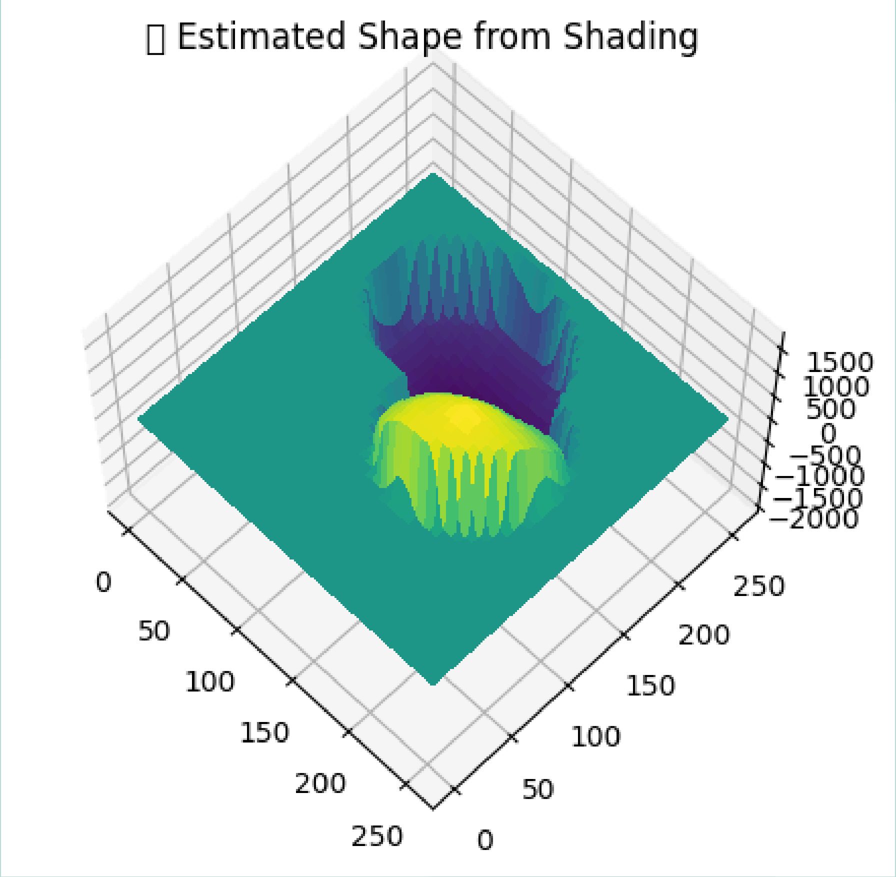
- Autonomous vehicles for obstacle detection and environment mapping.
- 3D scanning and modeling in engineering, gaming, and medical imaging.
- Augmented and virtual reality systems for immersive spatial experiences.

METHODS FOR 3D VISION

Input Image



Gradient image



Estimate Shape from Shading

FACE RECOGNITION USING EIGENFACES (PCA + KNN)

Definition

- Face Recognition is a biometric method that identifies individuals based on their unique facial features using image processing and machine learning techniques.

Purpose

- To design a real-time system that can automatically detect and recognize faces from a live webcam feed using Eigenfaces (PCA) and KNN classifier.

Concept Used

- PCA (Principal Component Analysis): Extracts key facial features (Eigenfaces) and reduces dimensionality.
- KNN (K-Nearest Neighbors): Classifies the detected face by comparing it with trained faces.

Applications

- Security and surveillance systems
- Automated attendance systems
- Mobile face unlock
- Access control and authentication

FACE RECOGNITION USING EIGENFACES (PCA + KNN)

Face Detection

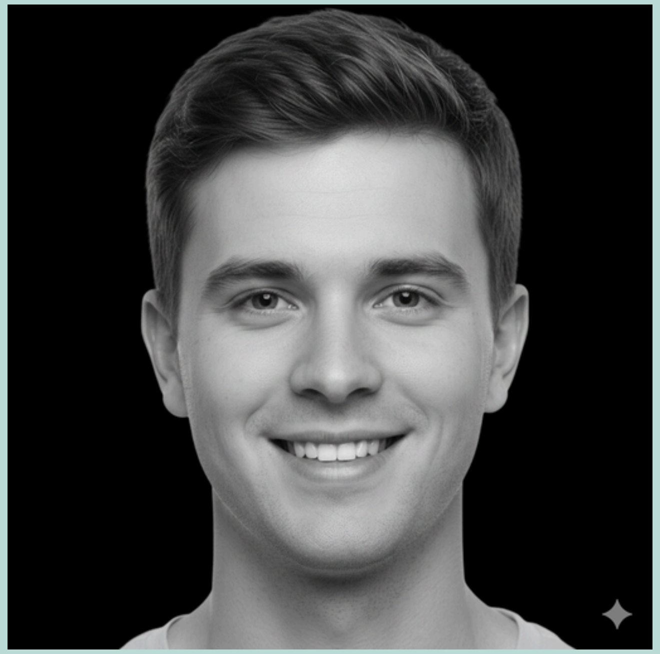
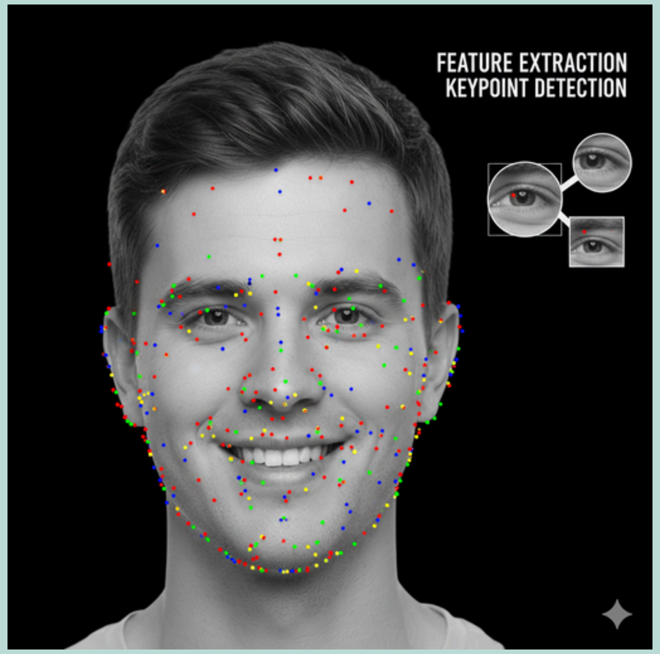
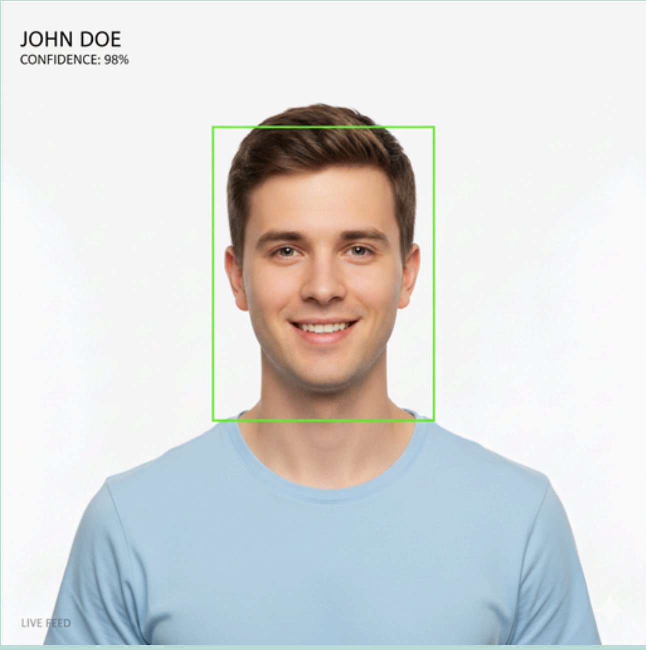


Image Preprocessing



Feature Extraction (PCA / Eigenfaces)



Output

GITHUB REPOSITORY

ComputerVisionProjectLink

Thank You